

Palladium Bank operates across five major Nigerian cities Lagos, Abuja, Kano, Port Harcourt, and Ibadan. The bank has accumulated 18 months of raw transaction data but currently runs reports directly on transaction logs. This approach is slow, inconsistent, and makes it difficult for leadership to answer critical business questions such as which customer segments generate the most fee income, how transaction behaviour varies by branch and channel, and which products drive deposits versus withdrawals. A dimensional model solves this by organising data into a structure optimised for fast, consistent, and flexible reporting.

The dimensional model chosen for Palladium Bank is a Star Schema. This design places one central fact table surrounded by dimension tables resembling a star shape. The Star Schema was chosen over other designs because it is simple to understand, fast to query, and works well with most business intelligence tools. It allows analysts to slice transaction data by customer tier, branch, product, channel, and time period without writing complex queries. This directly addresses the bank's need for faster and more consistent reporting.

The central fact table is named fact_transactions. The grain of this table is one row per transaction meaning each individual bank transaction is recorded as a single row. This was chosen because the bank's key business questions all revolve around individual transactions their amounts, types, and timing. The fact table stores measurable values including the transaction amount and balance after transaction. It also stores the transaction type as a degenerate dimension meaning it is a descriptive attribute stored directly in the fact table rather than in a separate dimension table, because transaction type alone does not justify its own dimension table.

Five dimension tables surround the fact table. The dimcustomer table stores customer details including their name and tier (Platinum, Gold, Silver, Standard). This enables customer behaviour analysis by segment. The dim_branch table stores branch name and state, enabling branch-level and regional reporting. The dim_product table stores product name and type (Account, Card, Loan, Digital, Savings), enabling product and channel performance analysis. The dim_channel table stores the transaction channel such as POS, ATM, Mobile App, or Branch,

enabling channel behaviour analysis. Finally, the dim_date table stores date details including day, month, quarter and year, enabling time-based analysis such as Month-on-Month, Quarter-on-Quarter and Year-on-Year comparisons.

Two dimensions were identified as likely to change over time and require Slowly Changing Dimension (SCD) strategies. The first is dim_customer a customer's tier can change over time as they increase or decrease their banking activity. SCD Type 2 was chosen for this dimension, meaning a new row is inserted whenever a customer's tier changes, preserving the full history of tier changes. This is important because the bank needs to track whether high-value customers are reducing activity over time. The second is dim_branch branch details such as name or state mapping may occasionally change. SCD Type 1 was chosen for this dimension, meaning the old value is simply overwritten with the new value, since historical branch name changes are not critical for analysis.

The recommended data loading approach for Palladium Bank is ETL (Extract, Transform, Load). This means data is first extracted from the raw transaction logs, transformed and cleaned before being loaded into the dimensional model. ETL was chosen over ELT because the bank's data requires significant transformation before it is useful including generating surrogate keys, standardising date formats, mapping customer tiers, and detecting SCD Type 2 changes in customer records. For the initial load, all 18 months of historical data will be loaded in the following order: dim_date first, then dim_customer, dim_branch, dim_product, dim_channel, and finally fact_transactions. This order ensures all foreign keys exist before the fact table is populated. For incremental loads, new transactions will be detected using the transaction date, duplicate transactions will be prevented using the txn_id as a unique identifier, and SCD Type 2 updates will be handled by checking if a customer's tier has changed since the last load. Four data quality checks were defined for this model. First, duplicate transaction check if a txn_id already exists in fact_transactions, the record is rejected to prevent double counting. Second, NULL amount check any transaction with a NULL or negative amount is flagged and held for review before loading. Third, referential integrity check any transaction

referencing a customer, branch, or product that does not exist in the dimension tables is rejected and logged for investigation. Fourth, date validity check any transaction date outside the expected 18-month range is flagged as potentially erroneous. For performance, the fact table is partitioned by year and month to speed up time-based queries. Indexes were created on all foreign key columns in the fact table to speed up joins between the fact and dimension tables. An aggregation table named `agg_monthly_branch_revenue` was also created to store pre-calculated monthly revenue by branch, making executive dashboards significantly faster to load.